Embedded 2D/3D Graphics IP Core
GSHARK Family

http://www.gshark.com

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Embedded 3D Graphics Target

Products of GSHARK Family

"GSHARK-TAKUMI Technology" family of OpenGL ES 2D/3D graphics accelerator IP core intended for use in embedded devices. Our products offer high performance, as well as small gate count and low-power consumption with built-in a high performance vertex processor and high-speed 2D sprite engine.

The GSHARK-TAKUMI Family currently consists of three products optimized gate-count and performance for each market.

Award - LSI of The Year 2003

GSHARK plus

Ultra Low power Consumption
Target Display Resolution : QCIF – QVGA
Suitable for Mobile system

GSHARK turbo

Enhanced performance & Quality & Function
Target Display Resolution : QVGA – VGA
Suitable for Advanced Mobile system

GSHARK2

High performance & High Quality & Function
Target Resolution : QVGA - XGA
Suitable for Advanced Mobile and Digital Consumer

Products of GSHARK Family

	GSHARK plus	GSHARK turbo	GSHARK2
3D Triangle Performance	0.5 M tri/sec	1.4M tri/sec	5M tri/sec
2D Performance	50M pixel	100M pixel	100M pixel
Color of frame buffer	16bit color	16/ 32bit color	16/ 32 bit color
Z buffer	16bit	16/ 24bit	16/ 24 bit
Texture unit number	1	2	2
2D/3D Hardware Acceleration	Yes	Yes	Yes
Vertex Processor (VP)	Yes	Yes	Yes
Programmable VP	No	Yes	Yes
2D Sprite Engine	Yes	Yes	Yes
BitBLT	Yes	Yes	Yes
OpenGL ES support	1.0	1.0 / 1.1	1.0 / 1.1
Gate count	220K	500K	850K

Concepts of GSHARK Family

- **Very Compact**

  GSHARK Family is carefully designed to keep gate count as small as possible. You can add high performance 2D/3D Graphics futures in small room of die area of your LSI.

- **Low Power Consumption**

  Our fine-tuned power management enabled less power consumption, and results longer battery life for mobile devices.

- **Easy to integrate with various software**

  All stages of 3D pipe line including geometry transform, lighting and generating 3D pixels are done by hardware.
  By storing 3D drawing commands in the external memory, consecutive commands are executed without CPU interaction.

Benefits

High performance of GSHARK Vertex Processor

GSHARK Family has hardware T&L engine. GSHARK turbo and GSHARK2 have a high performance programmable vertex processor. Programablity of 3D pipeline allows extending 3D futures and performance tuning.

Low memory band width

GSHARK Family uses optimized rasterizing algorithm, pixel cashing structure and hierarchical Z buffer functions, and resulting lower external memory band width.

Contribute to Low cost products

The GSHARK Family's very low gate-count contributes to lower cost SOC. and also lower power consumption.

Benefits

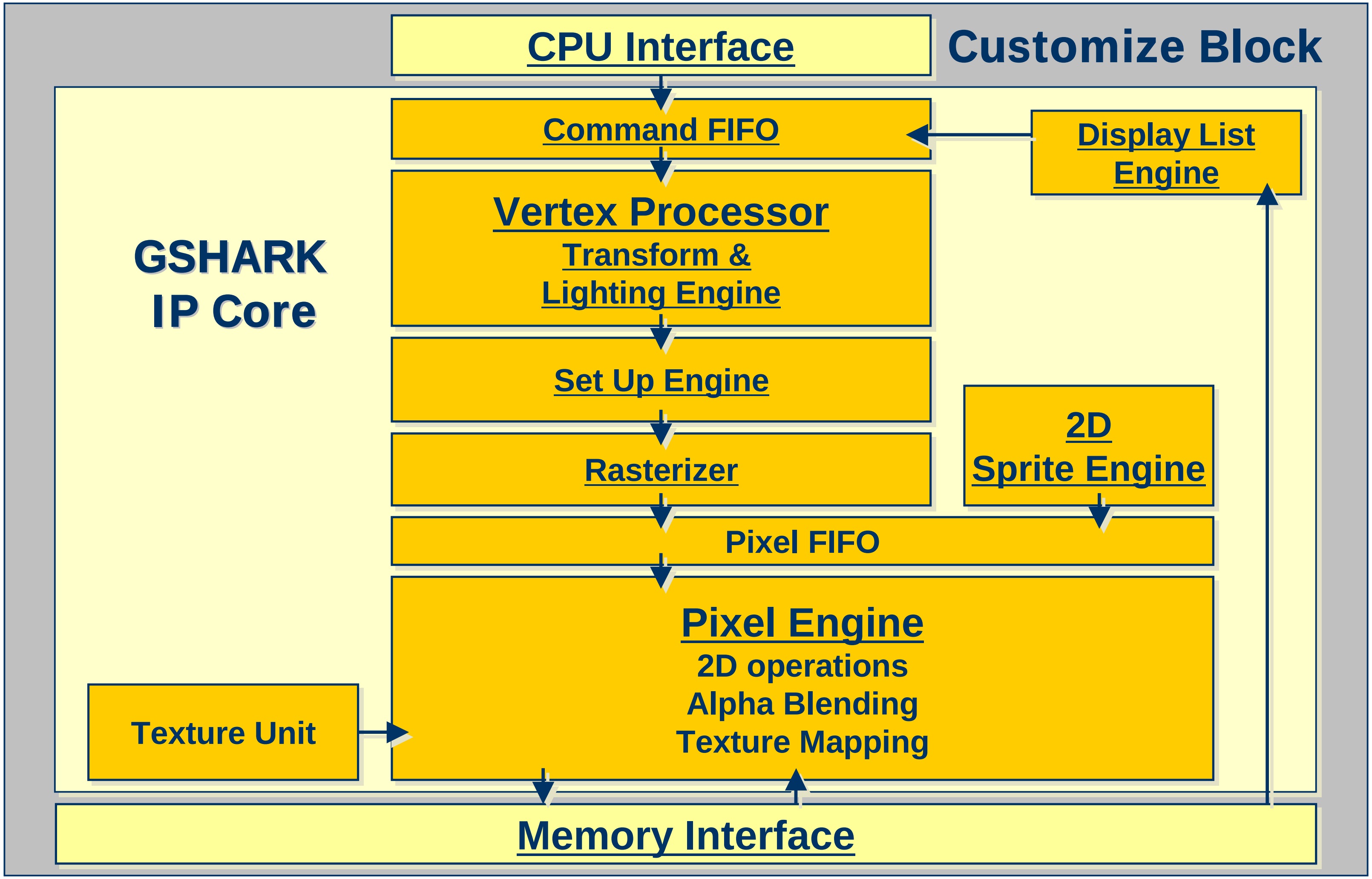
Unique function of 2D Sprite Engine

2D Sprite Engine has various 2D sprite function, such as scaling and rotation and transformation with Z and stencil and blending. GSHARK Family also has standard 2D function such as BitBLT and line draw and rectangle fill.

Easy integration for various SOC

GSHARK Family is delivered as a synthesizable soft IP. You can use any process of ASIC. The memory interface and CPU interface blocks are provided as customizable RTL code so that you can implement GSHARK core in your bus and memory design. Verified core design and API (OpenGL ES) ensure to reduce Time-to-Market.

GSHARK Block Diagram



GSHARKplus

GSHARKplus is one of the smallest OpenGL® ES HW 3D graphics Accelerator in worldwide. GSHARKplus has the 3D function and performance required the embedded mobile system.

Feature:

24bit floating point processor
Shading Mode: Flat, Smooth, Toon
Texture Filtering: Point, Bi-Linear
Texture Map Mode: Perspective Correct texture, Environment Mapping, MipMap, Indexed Color texture
Lighting: Point/ Spot/ Parallel, Ambient, Diffuse, Specular
Anti-aliasing: Full Scene, Line
Fog, Alpha-blending, Texture lighting
Display List Engine
2D: BitBLTs (ROP256), Point/Line/Fill Rect. Drawing
2D Sprite Engine: Rotation, Scaling, Transformation, Mesh drawing..
Frame buffer format: 16bit color (RGB565, RGBA5551)

GSHARKturbo

GSHARK-Turbo is OpenGL® ES1.1supported HW 3D graphics Accelerator and aiming mid range performance. Still small size but executes high performance for Advanced Mobile System and Digital Consumer Market.

Feature:

IEEE single precision 32bit floating point Computation (T&L)
Programmable Vertex processor
Shading Mode: Flat, Smooth, Toon
Texture Map Filtering: Point, Bi-Linear, Tri-Linear
Texture Map Mode: Perspective Correct, Multiple Texturing,
BumpMap, MipMap, Indexed Color, YUV Texture
Anti-aliasing: Full Scene, Line
Fog, Alpha blending, Texture lighting
Open GL Compatible: T&L, Alpha Blending…
Bit BLT(ROP256), Point/ Line/ Fill Rect. Drawing,
Display List Engine, Vertex Array Engine
2D: BitBLTs (ROP256), Point/Line/Fill Rect. Drawing
2D Sprite Engine: Rotation, Scaling, Transformation, Mesh drawing..
Frame buffer format: 16/32bit color
(RGBA8888, RGB565, RGBA5551,4444)

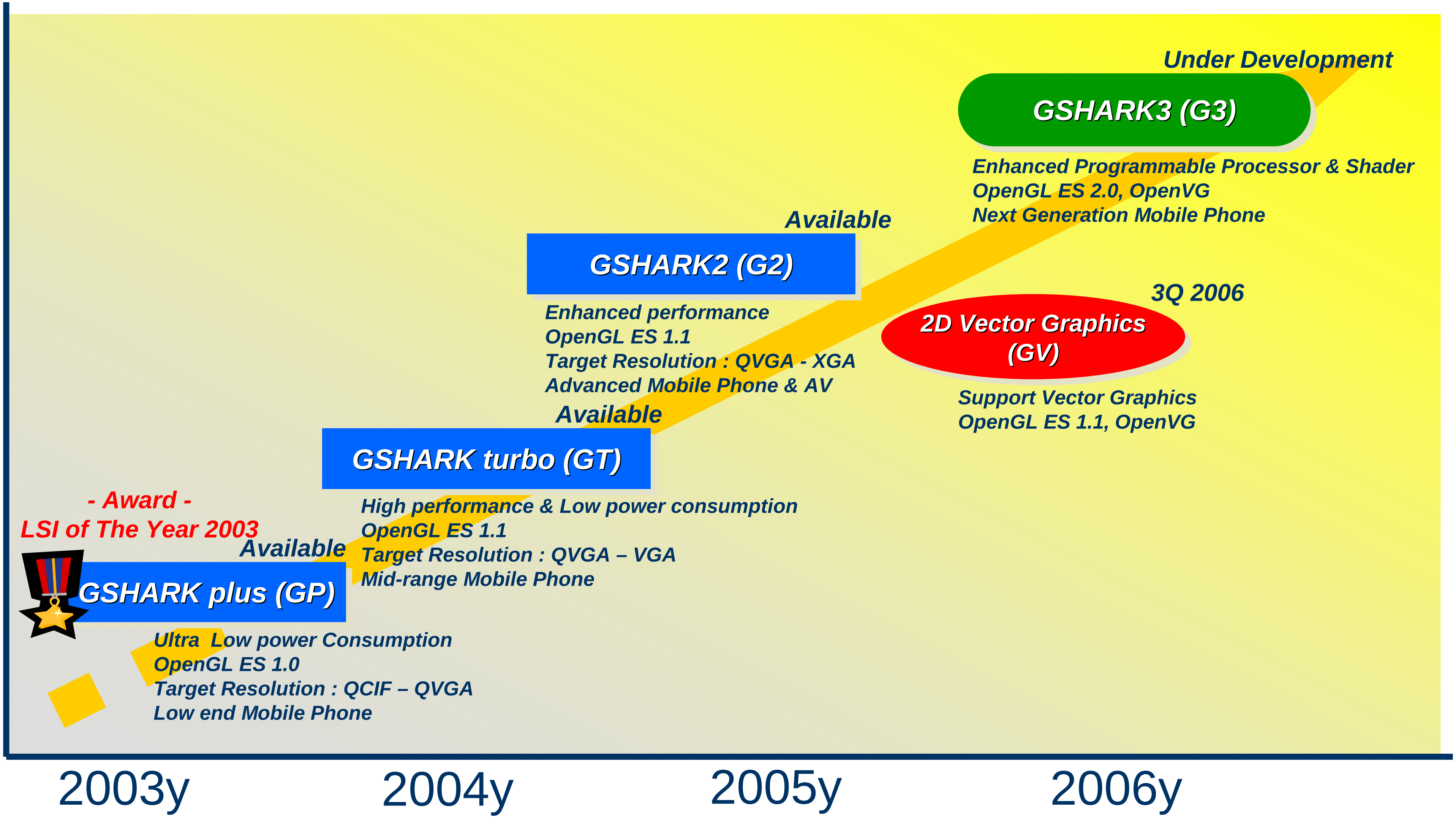
GSHARK2

GSHARK2 is Brand New Generation of GSHARK Family. Developed to responds to technological progress for Embedded System and globalization.
Brings you Highest Performance of Graphics in the market.

Feature:

IEEE single precision 32bit floating point Computation (T&L)
Programmable Vertex processor
Shading Mode: Flat, Smooth, Toon
Texture Map Filtering: Point, Bi-Linear, Tri-Linear
Texture Map Mode: Perspective Correct, Multiple Texturing,
BumpMap, MipMap, Indexed Color, YUV Texture
Anti-aliasing: Full Scene, Line
Fog, Alpha blending, Texture lighting
Open GL Compatible: T&L, Alpha Blending…
Bit BLT(ROP256), Point/ Line/ Fill Rect. Drawing,
Display List Engine, Vertex Array Engine
2D: BitBLTs (ROP256), Point/Line/Fill Rect. Drawing
2D Sprite Engine: Rotation, Scaling, Transformation, Mesh drawing..
Frame buffer format: 16/32bit color
(RGBA8888, RGB565, RGBA5551,4444)

GSHARK Family Road Map



Support Software

Supports API Library;

OpenGL ES 1.0 : ***GSHARK plus, turbo, 2***

OpenGL ES 1.1 : ***GSHARK turbo, 2***

CGL: ***GSHARK low-level API***

GSHARK plus ***has passed a Conformance Test and was approved as Hardware 3D accelerator based on*** ***OpenGLES 1.0*** ***from Khronos group(www.khronos.org) on Oct. 2004.***

GSHARK turbo ***has passed a Conformance Test and was approved as Hardware 3D accelerator based on*** ***OpenGLES 1.1*** ***from Khronos group(www.khronos.org) on Sep. 2005.***

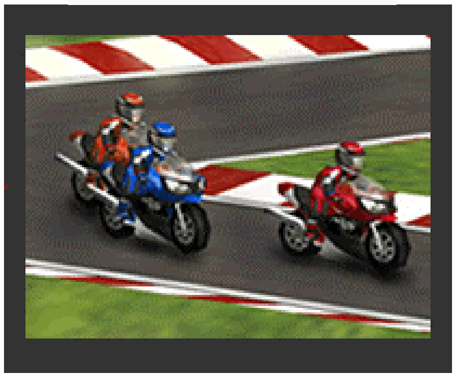
Software Development Supports; (option)

Verification Board: ***GT Core implemented in FPGA***

Partnership

Game Engine

HI Corporation



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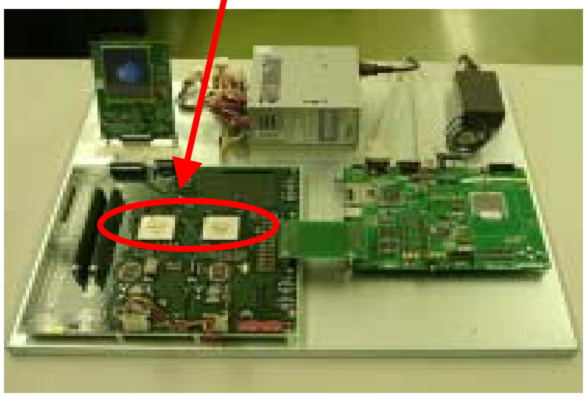
Demonstrations are ready for both game engines on the GSHARK Reference Board, accelerated by GSHARK IP core.

Reference Board

GSHARK plus Real Chip
Evaluation chip 80MHz

Demonstration Board

IP core is implemented in FPGA



Verification Board

It is possible to develop various 3D application software on the GSHARK verification Board with OpenGL ES Library and driver on Linux-OS.

Software Structure

Application
- C/C++ Application
- Java Application

Middleware
- 3D Middleware Engines
- Game Engines
- JSR184 library
- MIDP
- CLDC/ CDC
- Java VM

OS
- OpenGL/ES library
- GSHARK Driver
- Window Manager
- Linux Kernel
- Drivers

Hardware
- ARM9
- GSHARK

Concurrent Design

We prepare IP design in 3 models (C-model, RTL-model and Evaluation board). Our development environment realizes seamless programming between 3 models. By using it, you can program own test vectors to verify IP core implementation of your design, and you are able to start application programming before fabricating the LSI.